

Directive 071

Release date: December 1, 2025

Effective date: February 2, 2026

Replaces previous edition issued February 8, 2023

Emergency Preparedness and Response

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1 Introduction

1.1 Purpose of This Directive

This directive sets out the requirements for emergency preparedness and response for operations regulated under the *Oil and Gas Conservation Act (OGCA)*, *Oil Sands Conservation Act (OSCA)*, *Pipeline Act*, *Coal Conservation Act (CCA)*, *Geothermal Resource Development Act (GRDA)*, and *Mineral Resources Development Act (MRDA)* unless explicitly excluded. The requirements of this directive apply to the **approval holder**'s operations regardless of the phase of the development's life cycle (i.e., not limited to active operations). See appendix 1 for the definition of **emergency** and the meaning of **preparedness** and **response**.

The requirements in this directive are in addition to any plans, measures, directions, or orders otherwise required or authorized by the Alberta Energy Regulator (AER) and legislative requirements and are intended to prevent, alleviate, control, or stop the destruction, loss, or damage to the environment and protect public safety, human health, the environment, and property.

The directive includes elements from clauses 4 to 7 of the latest edition of the CSA Group standard [CSA Z246.2: Emergency Preparedness and Response for Petroleum and Natural Gas Industry Systems](#).

This directive **does not** address emergency preparedness and response requirements for dams and canals under the *Alberta Dam and Canal Safety Directive*, occupational health and safety in the *Occupational Health and Safety Code*, nor the security management program under the *Security Management for Critical Infrastructure Regulation*.

1.1.1 Emergency Management Program

This directive applies to approval holders and those developing, implementing, and maintaining the approval holders' **emergency management program (EMP)**, which includes the following elements:

- general elements (section 2.1), such as **senior management** sign-off of the "Emergency Management Program Declaration Form," documentation and document control, and record management
- program evaluation, continual improvement, and senior management review (section 2.2)
- management of change (section 2.3)
- **hazard** identification and consequence analysis (section 3)
- emergency preparedness involving the **public** and the **appropriate authority** (sections 4 and 10)

- **emergency response plan (ERP)** preparation and contents (sections 5, 11, and 12)
- competency and training (section 6)
- exercises (section 7)
- **incident** management (sections 8 and 13)
- emergency planning and **emergency response zones** for **hydrogen sulphide (H₂S)** and high-vapour-pressure (HVP) products (section 9)

Approval holders are required to manage their EMPs using a risk-based approach. This directive refers to the hazards associated with the approval holders' operations that require emergency preparedness to protect the public and the environment. EMPs are subject to AER audits to verify compliance and evaluate performance.

1.1.2 Emergency Response Plans

The ERP is a critical component of the approval holder's EMP. There are two types of ERPs:

- A **corporate ERP** is required for all operations and approval holders within the scope of this directive.
- An **operation-specific ERP** is an additional ERP required only for operations involving hazards associated with H₂S and HVP products; it requires AER approval.

"ERP" refers to both the corporate ERP and any operation-specific ERP.

1.2 How to Use This Directive

This directive applies to operations of different sizes, types, and hazards; therefore, the level of detail required for each program or plan will vary from one operation to another. The details of the approval holders' programs and plans must reflect the scale and complexity of their operations and the risks presented to the public and the environment. Depending on the scale and potential risk of the approval holder's operation, revisions or updates to existing programs or plans may be required under this directive.

Sections 2 through 8 of the directive include overarching emergency preparedness and response requirements (i.e., the EMP) applicable to all operations and approval holders within the scope of this directive. These sections require approval holders to follow a risk-based approach to emergency preparedness and response by considering hazards related to their operations and potential emergency scenarios arising from these hazards identified through hazard identification and consequence analysis (section 3). A corporate ERP is required for all operations and approval holders within the scope of this directive. Approval holders must communicate with the AER during an incident, in accordance with section 8.

Sections 9 through 13 of the directive include specific emergency preparedness and response requirements for approval holders whose wells, pipelines, and facilities (including **cavern** storage) contain H₂S or HVP products. An operation-specific ERP is required for operations involving H₂S or HVP products and associated hazards, which the approval holder must submit to the AER for approval (see section 11).

[*Manual 026: Emergency Preparedness and Response Guide*](#) provides information to support the approval holder to develop, implement, and maintain their EMP.

See the directive [landing page](#) for information on *Directive 071* forms.

Some terms are defined in text; other defined terms are set in **boldface** at first use, and the definitions are provided in appendix 1.

1.3 AER Requirements

In this directive, the term “must” indicates a requirement, while terms such as “should,” “recommends,” and “expects” indicate a recommended practice.

Each AER requirement that is unique to this directive is numbered.

Information on compliance and enforcement can be found on the AER website.

1.4 What's New in This Edition

The following changes have been made to this directive:

- The directive’s scope has expanded as follows:
 - Includes all operations under the *OGCA*, *OSCA*, *Pipeline Act*, *CCA*, *GRDA*, and *MRDA*, and not just wells and facilities as defined in the *OGCA* and pipelines defined in the *Pipeline Act*. Operations refer to any operation regardless of the phase of the development’s life cycle (i.e., not limited to active operations).
 - Includes applicable hazards that could result in an emergency requiring public safety and environmental protection measures, not just those from H₂S and HVP products.
- The term “duty holder” is replaced with “approval holder,” and “AER-approved ERP” with “operation-specific ERP.” These changes align with the revised language in the energy enactments.
- The directive now requires all approval holders to have an EMP.
- The directive now requires approval holders to perform hazard identification and consequence analysis to identify potential emergency scenarios associated with their operations and to consider the potential consequences for both public and environmental receptors.

- The public and the appropriate authority emergency preparedness requirements (sections 4 and 10) were revised to promote efficiency and reflect feedback from the public, appropriate authorities, and the industry.
- Spill contingency plan requirements are now in section 5 (Preparation of Emergency Response Plans), and spill response equipment training requirements are now in section 6 (Training).
- Section 7 (Exercises) requirements were revised to strengthen the effectiveness of exercises and promote continual improvement.
- The assessment matrix for **incident classification** in appendix 3 was updated to improve consistency in communication and response.
- The following forms were added: “Emergency Management Program Declaration Form” (section 2.1), “Emergency Management Program Self-Assessment Reporting Form” (section 2.2), and “Professional Declaration for Hazard Identification and Consequence Analysis” (section 3). The forms are available on the *Directive 071* [landing page](#).

2 Emergency Management Overview

The AER’s regulatory framework aims to protect public safety and minimize environmental effects through safe and responsible resource development. The AER’s regulatory instruments embody this aim and support the phases of the emergency management cycle: prevention and mitigation, preparedness, response, and recovery. *Directive 071* focuses on the preparedness and response phases of the management cycle.

The approval holder may develop, implement, and maintain their EMP as a standalone program or integrate it with other management programs. See section 2 of *Manual 026* for guidance on integrating EMP elements with an existing management program.

- 1) The approval holder must develop, implement, and maintain an EMP that, to the satisfaction of the AER, protects the public and the environment.
- 2) The approval holder must use the **incident command system** (ICS) to manage command and coordination of the emergency, as well as define the ICS roles and responsibilities with appropriate authorities during a response.

2.1 Emergency Management Program General Requirements

2.1.1 Management Commitment and Accountability

- 3) The approval holder must sign the “Emergency Management Program Declaration Form” and submit it to the AER with its corporate ERP via the designated information submission system. Then, the form must be resubmitted to the AER every 12 months thereafter. Form resubmission may be part of the corporate ERP update described in requirement 19.

The form is available on the directive [landing page](#).

2.1.2 EMP Performance Standards

- 4) The approval holder must set performance standards to ensure that EMP activities are conducted in a manner that protects the public and the environment.

2.1.3 EMP Program Coordinator

- 5) The approval holder must appoint a **program coordinator** for EMP development, implementation, evaluation, improvement, revision, and maintenance.

2.1.4 Documentation and Document Control

- 6) The approval holder must have the following:
 - a) a process for reviewing, revising, approving, controlling, and reissuing documents
 - b) schedules for reviews and revisions of the documents
 - c) EMP supporting documents, including the following:
 - i) EMP policies, roles and responsibilities, organizational structure, process, and procedures as required under sections 2.2 and 2.3
 - ii) ERPs as required under sections 5, 11, and 12
 - iii) a training plan as required under section 6
 - iv) an exercise plan as required under section 7

2.1.5 Record Management

- 7) The approval holder must
 - a) maintain records of its activities and decisions related to the EMP and
 - b) have a records management process for records identification, storage, protection, retrieval, retention, and disposition.

- 8) The approval holder must have a process for recording the following:
 - a) incidents (information gathered during and following an incident)
 - b) ERP maintenance
 - c) all records of public engagement, including individuals contacted, documented attempts of contact, and any outstanding issues yet to be resolved
 - d) training, meetings, and exercises:
 - i) records of staff training
 - ii) a record of the training completed within 60 days of any spill response equipment training
 - iii) a record of exercise results completed within 60 days of an exercise that includes
 - type of exercise held
 - exercise scope and objectives
 - persons involved
 - exercise outcome (i.e., whether objectives were achieved)
 - lessons learned
 - an action plan with timelines
 - iv) records of critical sour meetings (section 11.7), such as meeting sign-in sheets, invitations, and minutes, which are to be made available to the AER on request

The approval holder should keep all records for at least five years. The AER may request the approval holder to provide records at any time (e.g., for a hearing or audit).

2.2 Emergency Management Program Evaluation and Continual Improvement

- 9) The approval holder must implement a process to evaluate and continually improve the EMP that includes
 - a) a defined scope and
 - b) a methodology to monitor and measure program performance at planned intervals.
- 10) The approval holder must gather information to monitor the performance of the EMP, which may include incident statistics and performance indicators relating to the EMP objectives.

- 11) The approval holder must conduct a self-assessment at least every five years to determine whether the EMP meets the requirements of *Directive 071* and, upon request, submit the self-assessment results to the AER using the “Emergency Management Program Self-Assessment Reporting Form.”

The form is available on the directive [landing page](#).

- 12) The approval holder must correct deficiencies, gaps, and limitations identified during an EMP evaluation, self-assessment, or management review within the planned intervals specified in this directive or as requested by the AER during a compliance review.

See section 2.2 of *Manual 026* for guidance regarding how approval holders can evaluate the performance of their management review process.

2.3 Management of Change of the Emergency Management Program

- 13) The approval holder must have a management of change process to identify, review, and approve EMP changes before their implementation and ensure that changes are analyzed, documented, and communicated to affected personnel. The approval holder’s management of change process must consider EMP changes that could affect the following:
 - a) the public and the environment
 - b) hazard identification or consequence analysis results
 - c) the approval holder’s emergency management structure
- 14) The approval holder’s management of change process must account for updating the EMP because of changes in AER regulatory requirements.

See section 2.1 of *Manual 026* for guidance regarding how approval holders can evaluate the performance of their management of change process.

3 Hazard Identification and Consequence Analysis

Hazards are potential sources of harm to people, the environment, or property. It is essential to identify hazards that could result in an emergency, as described in appendix 3, analyze their effects (consequences), and develop effective emergency preparedness and response strategies accordingly.

The level of detail in the hazard identification and consequence analysis, as well as the competency and qualifications of the person conducting the assessment, should be proportionate to the scale and complexity of the operations and appropriate to the nature of the hazard. The approval holder may choose whether to document or store the EMP separately from its ERPs. Section 3 of *Manual 026* provides guidance on hazard identification and consequence analysis.

- 15) The approval holder must take the following actions concerning hazards associated with its operations, regardless of the life-cycle stage:
- a) Identify hazards and the potential emergency scenarios arising from these hazards, ranging from the worst-case to likelier but less severe scenarios, that could result in moderate, major, or catastrophic consequences described in appendix 3. These scenarios include
 - i) uncontrolled, unplanned, or accidental release of a **substance** or a mixture of substances;
 - ii) failure of structures or geological features used for containment or from excavations;
 - iii) natural hazard events affecting operational safety; and
 - iv) any other emergency scenario within the scope of this directive (section 1.1) specific to the approval holder's operations.
 - b) Identify and analyze the potential consequences to the public and the environment resulting from the identified potential emergency scenarios, focusing on the worst-case and worst credible scenarios.

See section 3 of *Manual 026* for examples of emergency scenarios that fall within or outside the scope of *Directive 071* and explanations of the worst-case and worst credible scenarios.

- 16) The approval holder must have a process for identifying, analyzing, and managing hazards that includes the following:
- a) appointing a person or company knowledgeable in the industry and discipline to identify hazards and analyze consequences
 - b) establishing and maintaining documentation of identified hazards and potential emergency scenarios
 - c) establishing and implementing a process for evaluating and managing the consequences associated with the identified hazards

See section 3 of *Manual 026* for guidance on documenting identified hazards and potential emergency scenarios.

- 17) Complete the "Professional Declaration for Hazard Identification and Consequence Analysis" form for hazards associated with the approval holder's operations that could result in catastrophic consequences to the public and the environment (see appendix 3) and operations involving H₂S and HVP products that require **emergency planning zone (EPZ)** determinations (see section 9). This form must be signed or stamped by a licensed professional or a permit holder registered with the Association of Professional Engineers and

Geoscientists of Alberta. The approval holder must retain this form and the corresponding documentation and make them available to the AER on request.

The form is available on the directive [landing page](#).

4 Emergency Preparedness Involving the Public and the Appropriate Authority

Ensuring that the public and the appropriate authority are aware of potential emergencies from approval holders' operations and their involvement in emergency preparedness and emergency response activities is a fundamental component of the approval holders' EMP.

The approval holder's emergency preparedness activities involving the public and the appropriate authority include

- establishing relationships with the public, appropriate authorities, and other industry operators;
- developing a one- or two-page quick guide to emergencies for corporate ERPs (section 5.2) and distributing it upon receiving a written request to the public or an appropriate authority;
- developing and distributing a public emergency preparedness information package (section 10.1.1) for operation-specific ERPs; and
- conducting emergency preparedness activities involving the public and local authorities for operation-specific ERPs (sections 10.1 and 10.2).

The level of involvement by the public or appropriate authorities in emergency preparedness and response may vary. Where the public or appropriate authorities have the capability and are willing to be involved in the approval holder's emergency preparedness and response activities, the approval holder must involve them in the process in accordance with this directive. Any appropriate authority that has indicated it wants to receive emergency preparedness and response information from the approval holder, as required by this directive, should contact the AER at Directive071@aer.ca if they are not receiving the information.

For more information regarding practices that establish relationships, including the use of **synergy groups**, see section 9 of *Manual 026*.

5 Preparation of Emergency Response Plans

- 18) The approval holder must use either a single ERP or a set of coordinated ERPs developed at a level of detail and complexity proportionate to its operations and that addresses the hazards and potential consequences of the emergency scenarios that its operations pose to the public and the environment, and includes
 - a) all identified potential emergency scenarios and
 - b) the procedures that will aid in effective emergency response.

- 19) The approval holder must review its corporate ERP and any operation-specific ERPs every 12 months, including incorporating any supplements described in section 11.2, or where changes to operations or emergency response personnel occur, immediately correct errors in the ERP upon discovery or when reported to them by the AER or other parties and submit the updated ERP to the AER via the designated information submission system.
- 20) When an operation-specific ERP is no longer required, the approval holder must inform all parties associated with the subject ERP that it is no longer in effect (see section 10.5.1 of *Manual 026* for details).
- 21) If the AER determines to conduct a hearing to consider an application, the approval holder must provide its relevant ERPs to the AER for review. Once the AER has determined that the approval holder's ERPs are **technically complete**, a notice of hearing may be issued.

5.1 Corporate Emergency Response Plan

- 22) The approval holder must develop its corporate ERP that includes the common components in section 5.3 and keep it current.
- 23) The approval holder must submit its corporate ERP to the AER via the designated information submission system as follows:
 - a) For approval holders with operations that started on or before the effective date of this edition of the directive, the submission must occur no later than the next scheduled 12-month update, unless otherwise specified by the AER in writing.
 - b) For approval holders that start operations after the effective date of this edition of the directive, the submission must occur before the start of operations.

Although corporate ERPs do not require AER approval, they are subject to review and audit by the AER.

5.2 Quick Guide to Emergencies

- 24) The approval holder must develop a one- or two-page quick guide to emergencies that includes the following:
 - a) a brief overview of the operations and the associated hazards
 - b) a geographic reference of the operations in the area
 - c) the active 24-hour emergency contact telephone number for the approval holder, the AER's Energy and Environmental Emergency 24-Hour Response Line (1-800-222-6514), and the telephone numbers for **local authority** contacts

- d) a summary of the emergency response procedures in place or a reference to other documents containing the procedures
 - e) information on measures in place to protect the public and the environment
- 25) The approval holder must provide a quick guide to emergencies to the public or local authorities on request.

5.3 Components Common to All Emergency Response Plans

This section sets out the requirements common to all ERPs (i.e., corporate ERPs and operation-specific ERPs).

See section 12 for requirements that apply to operation-specific ERPs.

5.3.1 Incident Detection and Confirmation Procedures

- 26) The approval holder's ERP must set out the following procedures for incident detection, notification, and confirmation:
- a) How incidents are detected.
 - b) How the approval holder is alerted of an incident.
 - c) How incidents are confirmed (e.g., a release).

5.3.2 Incident Classification and Protection Measures

The AER has developed an assessment matrix to classify incidents so the approval holder can consistently communicate with the appropriate authority and other industry operators. There are four levels of incident classification (in ascending order of risk and severity): **alert**, **level 1 emergency**, **level 2 emergency**, and **level 3 emergency**. For more information, see appendix 3.

- 27) The approval holder must include the AER's incident classification matrix in its ERP.
- 28) The approval holder must define and set out in the ERP the appropriate actions for each incident level, including measures to protect the public and the environment related to the potential emergency scenarios identified and potential consequences analyzed in accordance with the requirements of section 3.

5.3.3 Communications Plan

The development and implementation of an effective communications plan is essential to incident response.

- 29) The approval holder's ERP must include a communications plan that includes

- a) procedures for contacting and maintaining communication with key approval holder personnel, appropriate authorities, support services, emergency social services, the public, the media, and other affected approval holders;
 - b) clearly defined roles and responsibilities to contact the AER and other responders identified in the ERP in an emergency;
 - c) procedures that will be implemented during an incident to contact and maintain communication with persons affected to keep them informed of the situation and actions being taken, including plans for communicating the implementation of **public protection measures** (section 13.4);
 - d) procedures to inform and update the media and to distribute factual messages to the public in an expeditious manner; and
 - e) procedures to downgrade an incident or stand down an incident.
- 30) The approval holder must have redundant methods of emergency communication and notification to ensure that those affected by an incident are informed.

5.3.4 Emergency Contact Phone Numbers

- 31) The approval holder's ERP must include the following:
- a) the 24-hour emergency phone number for the approval holder and the Energy and Environmental Emergency 24-Hour Response Line phone number (1-800-222-6514) at the front of the ERP for easy access
 - b) a telephone list of internal incident response personnel
 - c) a telephone list of external emergency support services that might be required during an incident (see section 4 of *Manual 026* for a listing of potential external support services)

5.3.5 Public and Environmental Protection Measures

- 32) The approval holder's ERP must include measures to protect the public and the environment appropriate for the emergency scenarios identified and potential consequences analyzed in accordance with the requirements of section 3.

5.3.6 Spill Response Procedures

The requirements in this section apply to approval holders with potential emergency scenarios involving uncontrolled, unplanned, or accidental release of the following substances onto land or water:

- oil, water, or unrefined product per the *Oil and Gas Conservation Rules*

- operating fluid, oilfield waste, water, or hydrocarbon per the *Geothermal Resources Development Rules*
- crude bitumen, oil sands product, condensate, salt water, or contaminated water per the *Oil Sands Conservation Rules*
- produced fluid, brine-hosted mineral resources, operating fluid, oilfield waste, water or hydrocarbons per the *Brine-Hosted Mineral Resources Development Rules*
- any flow of oil or water encountered during mining per the *Rock-Hosted Mineral Resources Development Rules*
- any flow of oil or water encountered during exploration or mining per the *Coal Conservation Rules*
- any substance escapes from a pipeline per the *Pipeline Act*

The requirements in this section do not apply to infrastructure or sites abandoned following AER requirements or to operations that produce, transport, or store less than two cubic metres per month of the above-listed substances.

An approval holder who is a member in good standing of a **spill cooperative** in Alberta covering the area where the approval holder operates is considered to meet requirement 33 by referencing the implementation procedures for the spill cooperative's spill contingency plan in its ERP. Also, an approval holder who is a member in good standing of a spill cooperative in Alberta covering the area where the approval holder operates is considered to meet requirement 36.

- 33) The approval holder's ERP must include the following to address the release of substances listed above onto land or water from any operations within the scope of this directive (e.g., well, **facility**, pipeline, mine, etc.) or any trucking associated with the operations:
- a) a description of initial emergency response procedures and actions with relevant contacts and services
 - b) an inventory of operations carrying the substances listed above that are near or crossing a **water body**
 - c) topographical maps showing designated spill control points (if applicable), access roads, **urban centres**, water bodies (i.e., streams, rivers, lakes), and water supply intakes for municipal and industrial operations, pipelines, wells, and facilities within the operating area
 - d) roles, responsibilities, and resources to manage the environmental protection response
 - e) inventory and location of response equipment

- f) containment and recovery procedures applicable to the type, volume, and nature of the production and the time of year
- 34) To be considered a member in good standing of a spill cooperative in Alberta covering the area where the approval holder operates, an approval holder must meet the obligations and intent of the cooperative.
- 35) The approval holder must deploy trained responders for spill response.
- 36) The approval holder must
 - a) purchase appropriate spill cleanup equipment, considering the type of operations and terrain in which the approval holder operates,
 - b) maintain the equipment in good working order, and
 - c) store the equipment where it may be required for immediate access.

5.3.7 Mutual Aid and Third-Party Contracts or Operations

The term “mutual aid agreement” or “mutual assistance agreement” includes cooperative agreements, partnership agreements, memoranda of understanding, and other terms commonly used to describe the sharing of resources.

See section 4 of *Manual 026* for details on mutual aid agreements.

- 37) Approval holders with **mutual aid understandings** with third parties must include a summary of these understandings in their ERP.
- 38) Approval holders with operations physically connected to another approval holder or whose assets are operated by a third-party contractor and have mutual aid understandings in place must include a bridging statement summary in their ERP that includes
 - a) the communication protocols between the approval holders during an incident and
 - b) a description of any shared emergency response procedures and support.

A bridging statement references the ERP of another approval holder operating an asset tied together or owning an asset tied together. A bridging statement also applies in cases where assets are transferred from one approval holder to another (i.e., temporarily until the approval holder can integrate the transferred assets into their ERP). A bridging statement may contain

- the terms of the agreement between the approval holders,
- emergency contacts,
- a 24-hour emergency number for each approval holder, and
- the communication and response protocol.

5.3.8 Incident Management Procedures and Personnel Responsibilities

- 39) The approval holder's ERP must include
 - a) procedures specific to its operations to manage and coordinate incident response and
 - b) roles and responsibilities during an emergency response for operational command personnel, incident support facilities, and any other first responders.

5.3.9 Registration and Support Centre Activation Procedures

- 40) The approval holder must include procedures in its ERP for selecting and activating a registration and support centre at a safe distance from the incident.

5.3.10 Emergency Response Equipment List

- 41) The approval holder's ERP must include a list of emergency response equipment relevant to its emergency response procedures that identifies the location, number, and type of equipment for use in incident response, including
 - a) communications equipment needed for emergency response personnel (the approval holder is responsible for ensuring that communications equipment is made available to response personnel),
 - b) roadblock kits,
 - c) spill cleanup equipment (if applicable),
 - d) on-site ignition equipment (if applicable), and
 - e) gas monitoring equipment (if applicable).
- 42) If the equipment is from a third party, the approval holder's ERP must include information regarding its location and the estimated time to the site.

5.3.11 Hazard Monitoring During an Incident

- 43) The approval holder's ERP must include procedures for ongoing monitoring of hazards throughout an incident.
- 44) If applicable, the approval holder's ERP must include protocols to support a decision to ignite a release, including emergency response procedures for immediate ignition and actions taken if a release occurs while personnel are on site.

5.3.12 ERP Distribution List

- 45) The approval holder's ERP must include a list of individuals and parties who received copies of the ERP.

5.4 Transfer of Licences, Approvals, and Amalgamations

This section applies to emergency preparedness and response requirements related to AER licences and approvals subject to purchase and sale agreements, including corporate sales or amalgamations, and the subsequent AER licence transfer and approval process.

For *Directive 071*, the transition period begins on the date of transfer for the subject licences or approvals or amalgamation and ends on the date the new approval holder receives the ERP approval. See section 4.3 of *Manual 026* for details on the transition period for licence transfers, approvals, or amalgamations.

- 46) The approval holder of record with the AER remains responsible for any ERPs associated with the subject sale for the transition period.

The prospective approval holder must follow the requirements specified in this directive for relevant changes resulting from the transfer of licences, approvals, or amalgamations, which may include updating the corporate ERP during its next 12-month ERP update cycle.

- 47) For assets contained within an operation-specific ERP, the prospective approval holder involved in the licence or approval transfer or amalgamation must submit a letter to Directive071@aer.ca notifying the AER of the incoming transfer within 30 days of the transfer of the licence or approval, as well as discuss a timeframe for submitting a new ERP or an amendment to an existing ERP. The notification letter must include the following information:
 - a) Confirmation of the 24-hour emergency contact number for the subject assets and the new approval holder's primary contact information.
 - b) The anticipated submission date for either a new ERP or an amendment to an existing ERP.
 - c) List the ERP registration numbers for all operation-specific ERPs in effect during the transition period. If none of the existing operation-specific ERPs will be used, the letter must provide the AER with a detailed interim plan that satisfies the ERP requirements for the transition period.

The AER will review the notification letter and advise the current or prospective approval holder of any concerns to be addressed.

- 48) When an operation-specific ERP is required, the new approval holder must
 - a) notify **residents** within the EPZ and any appropriate authorities of the change in ownership and
 - b) develop a new operation-specific ERP.

6 Training

- 49) Training is a key component of the EMP. See section 5 of *Manual 026* for details on training plans. The approval holder must ensure that personnel participating in the EMP, including response personnel, are competent, have a combination of the required education and experience, and have completed the training specified in the approval holder's training program for their assigned roles and duties.
- 50) The approval holder must establish and implement a training program, including training plans.

6.1 Emergency Response Plan Procedures Training

- 51) The approval holder must provide ERP training that includes the following:
 - a) all components of the plan, and the supporting tools and technology
 - b) roles and responsibilities during an incident
 - c) measures to protect the public and the environment used during an incident
 - d) planned communication protocol

6.2 Spill Response Equipment Training

An approval holder who is a member in good standing of a spill cooperative in Alberta covering the area where the approval holder operates is considered to meet requirement 52 if it meets either of the following conditions:

- The approval holder has participated (and was appropriately represented by spill response personnel) in at least one spill cooperative annual deployment exercise in the area of its operations (attendance at an exercise outside the area where the approval holder operates is also acceptable).
 - Area representatives have completed a spill response course (excluding online courses) or an on-scene spill commander course from a recognized training institution (this option cannot be used in consecutive years by the approval holder).
- 52) The approval holder must conduct annual training in the area of its operation, involving the same field supervisory personnel and using the spill response equipment, and must notify the AER via the designated information submission system at least 30 days in advance of the training and provide the following information:
 - a) The nature of the training exercise, the exercise date, and the legal land description of the land on which it will be conducted.

- b) A map showing the general topography, location of and access routes to the deployment area, and the location of any municipal water intakes within three kilometres of the deployment area.
- c) If the spill training exercise involves using a spill material, the approval holder must identify the proposed material and planned volume.
- d) Comments on the public use of the area, the collection and disposal of garbage, and a statement indicating the extent, if any, of anticipated surface disturbance to stream banks or shorelines at the test site.
- e) Confirmation of the training exercise occurring on private land, and whether the landowner agrees to the exercise proceeding at the proposed site.

7 Exercises

An exercise is a simulated emergency where responders perform their duties as they would in a real emergency. Exercises can validate ERPs and procedures, as well as practice the approval holder's emergency response capabilities.

In addition to the training activities required in section 6 of this directive, exercises will identify any gaps in response and create practical experience in response roles. See section 6 of *Manual 026* for guidance on exercises.

53) The approval holder must conduct the following exercises:

- a) an annual **tabletop** or **functional exercise** for a corporate ERP;
- b) an annual tabletop or functional exercise for each operation-specific ERP except in the year when holding a **major exercise**; and
- c) a major exercise conducted at all levels of the organization, including senior management, as follows:
 - i) If the approval holder has an operation-specific ERP, hold the exercise in the first year of implementation and every three years thereafter for each operation-specific ERP.
 - ii) If the approval holder does not have an operation-specific ERP, hold the exercise every five years for a corporate ERP.

54) The approval holder must include in its exercises any individuals acting on behalf of the operator having a defined role in incident management and ensure that exercises test the following:

- a) a range of potential emergency scenarios, incorporating variations to address the emergency scenarios identified in accordance with section 3;
 - b) public safety and environmental protection procedures identified in an ERP; and
 - c) mutual aid agreements or cooperatives, where applicable.
- 55) If multiple approval holders choose to conduct a joint exercise as part of a synergy or mutual aid group, the relevant local authority and the Alberta Emergency Management Agency must be invited to participate, and the exercise must
- a) be a major exercise,
 - b) test mutual aid procedures outlined in each approval holder's ERP, and
 - c) ensure all involved approval holders are represented in command or general staff positions.

Approval holders may satisfy multijurisdictional exercise requirements in a single exercise, provided that the AER's requirements are met (i.e., where a multijurisdictional response is necessary under the tested ERP).

Where approval holders have multiple ERPs with the same field supervisory response personnel and infrastructure, the ERPs may be tested simultaneously through a single exercise.

Multiple approval holders can jointly host an exercise as part of a synergy or mutual aid group.

- 56) The approval holder must
- a) notify the AER of a tabletop or major exercise, and submit the notification to the AER via the designated information submission system; and
 - b) invite the local authorities identified in the ERP no later than 30 days before a scheduled major exercise and include the emergency scenario details in the written invitation provided to the local authorities.

See *Manual 026* for guidance on exercise notification and invitation expectations, including procedures for handling changes to a planned exercise.

8 Incident Management

8.1 Initial Response

- 57) The approval holder must post signage with its active 24-hour emergency telephone number at the primary entrance to all operations regulated under the *OGCA*, *OSCA*, *GRDA*, *MRDA*, and the *Pipeline Act* unless explicitly excluded.

- 58) The approval holder must ensure that a call to its 24-hour emergency telephone number initiates immediate action.
- 59) The approval holder must immediately respond if notified of an incident by an alarm or other type of incident notification.

8.2 Classifying an Incident

- 60) The approval holder must use the assessment matrix in appendix 3 to classify the incident.

See section 7.1 of *Manual 026* for the recommended actions for each incident level.

- 61) The approval holder must contact the AER immediately through the Energy and Environmental Emergency 24-Hour Response Line phone number (1-800-222-6514) after it has activated internal response resources to confirm the incident level and convey the incident specifics. Ongoing communication with the AER must be maintained throughout the incident unless otherwise directed by the AER.

The approval holder makes the initial determination of the incident level classification so that it can immediately communicate it and activate its internal response resources as necessary. The AER may verify the information from the approval holder and reclassify the incident level accordingly.

8.3 Measures to Protect the Public and the Environmental

- 62) The approval holder must ensure that measures to protect the public and the environment are carried out in accordance with the ERP, covering the relevant operational areas and emergency scenarios.

8.4 Communication During an Incident

- 63) If an emergency extends beyond the approval holder's site and could potentially affect the public or the environment, or if the approval holder has contacted the public or the media about the emergency, the approval holder must
 - a) notify the local authorities listed in the ERP and
 - b) mobilize its support services required to assist with the initial response.

- 64) The approval holder must notify the public (residents, stakeholders, and Indigenous groups) affected by the emergency.

See requirement 61 for communication with the AER during an incident.

8.5 Hazard Monitoring During an Incident

- 65) The approval holder must ensure hazard monitoring is in place during an incident in accordance with the ERP (see section 5.3.11).

8.6 Access Control

- 66) The approval holder must ensure that manned access control is in place to restrict unauthorized entry into the response areas during an incident.

See section 7.3 of *Manual 026* for details on access control.

8.7 Equipment Requirements

- 67) The approval holder must ensure that the equipment identified in the ERP is available and at the location specified in the ERP before conducting any operations.
- 68) The approval holder must ensure that the equipment identified in the ERP for drilling and completion operations for **critical sour wells** is available at the location specified in the ERP before entering the critical sour zone.

In this directive, critical sour zone and sour zone refer to subsurface geological formations.

- 69) The approval holder must ensure that equipment is operational and that documentation is available to verify testing and calibration requirements.

8.8 Downgrading an Emergency Level

- 70) Once the consequences and potential for de-escalation of the emergency are re-evaluated, the approval holder must decide in consultation with the AER whether to downgrade the emergency level.
- 71) The approval holder must keep all notified persons, evacuated persons, agencies, and the media informed of the incident status.

9 H₂S and HVP Products – Emergency Planning Zones and Emergency Response Zones

This section sets out requirements for determining EPZs and emergency response zones for operations involving H₂S or HVP products.

- 72) The approval holder must determine the EPZ and emergency response zones for wells, pipelines, and facilities (including cavern storage) containing H₂S or HVP products or both.

9.1 Emergency Planning Zone

An EPZ is a geographic area around wells, pipelines, or facilities where the presence of H₂S or HVP products requires specific emergency preparedness by the approval holder. The EPZ is used for emergency preparedness planning and does not necessarily define the response area, nor is it assumed to be a dividing line between the approval holder's response and that provided by other emergency responders.

The AER may audit the inputs and methodologies used for calculating EPZs.

- 73) The approval holder must ensure that the size and shape of the EPZ reflect
- a) information gathered during the public preparedness program (see section 10) and
 - b) population density, topography, access and egress routes, and other features affecting timely emergency response.

The hazard location is the reference point for EPZ distance measurements, as indicated in the following examples:

- The EPZ for a well is measured from the wellhead.
 - The EPZ for a pipeline is measured from the centreline of the pipeline.
 - The EPZ for a facility handling or processing sour fluids is the largest EPZ of any pipeline entering or leaving the facility, measured outward in all directions from the facility lease boundary. However, if the facility has a **sour well**, sour water disposal well, or acid gas disposal well on site, the EPZ for the well may determine the size of the facility EPZ.
- 74) Before filing an ERP, the approval holder must use the AER-designated H₂S model, as specified in *Manual 026*, to calculate the size of the EPZ for operations with an H₂S concentration of 0.1 moles per kilomole (mol/kmol) or more.
- 75) When submitting its ERP application, the approval holder must upload a copy of the comma-separated values (CSV) batch export file from the AER-designated H₂S model to the designated information submission system.

The AER may audit the inputs and methodologies used for calculating EPZs. See section 8 of *Manual 026* for details on the AER-designated H₂S model.

- 76) During any operation involving H₂S or HVP products, the approval holder must ensure that on-site supervisory personnel are aware of the size of the EPZ.

9.2 Overlapping Emergency Planning Zones

- 77) Where two or more wells are close to one another such that their EPZs overlap, approval holders conducting sour drilling or completion operations, wellhead-off **workovers**, or **well servicing** operations in any formation containing H₂S that is open to the wellbore must
- a) jointly confirm whether the EPZs overlap and if sour operations are scheduled to occur simultaneously,
 - b) review and modify ERPs as required (e.g., communication protocol changes),
 - c) advise the AER in writing before conducting sour operations, and

- d) establish, maintain, and document communication among the approval holders.

Approval holders may ask the AER for a revised EPZ based on a technical evaluation of the potential release rate. The technical evaluation should consider sour zones isolated behind casing and a valid analysis of the H₂S concentration from the zones open to the surface.

During drilling operations for critical sour wells with overlapping EPZs, once the first well penetrates the critical sour zone porosity, maintains well control, and establishes that there is no sour gas cap, the second well may proceed to penetrate the critical zone, which could be a standalone zone or a combination of zones that makes the well a critical sour well.

9.3 Emergency Response Zones

There are two zones in which an approval holder's emergency response resources will be focused during an H₂S release emergency: the **initial isolation zone (IIZ)** and the **protective action zone (PAZ)** (collectively referred to as emergency response zones).

9.3.1 Initial Isolation Zone

The IIZ is the area near a hazardous substance release where indoor sheltering may only provide temporary protection due to the proximity of the release.

- 78) If safe to do so, the approval holder must attempt to evacuate the public from the IIZ.

9.3.2 Protective Action Zone

The PAZ is the area downwind of a hazardous substance release where outdoor pollutant concentrations may result in life-threatening or seriously irreversible health effects on the public and emergency responders.

Immediately following a release of H₂S or HVP products, the approximate size and shape of the PAZ can be determined using the conditions at the time of the incident (see figure 1). Once monitoring equipment is deployed, the actual size and shape of the PAZ can be established based on the monitored conditions. The PAZ size, shape, and compass orientation may change over time based on monitored conditions.

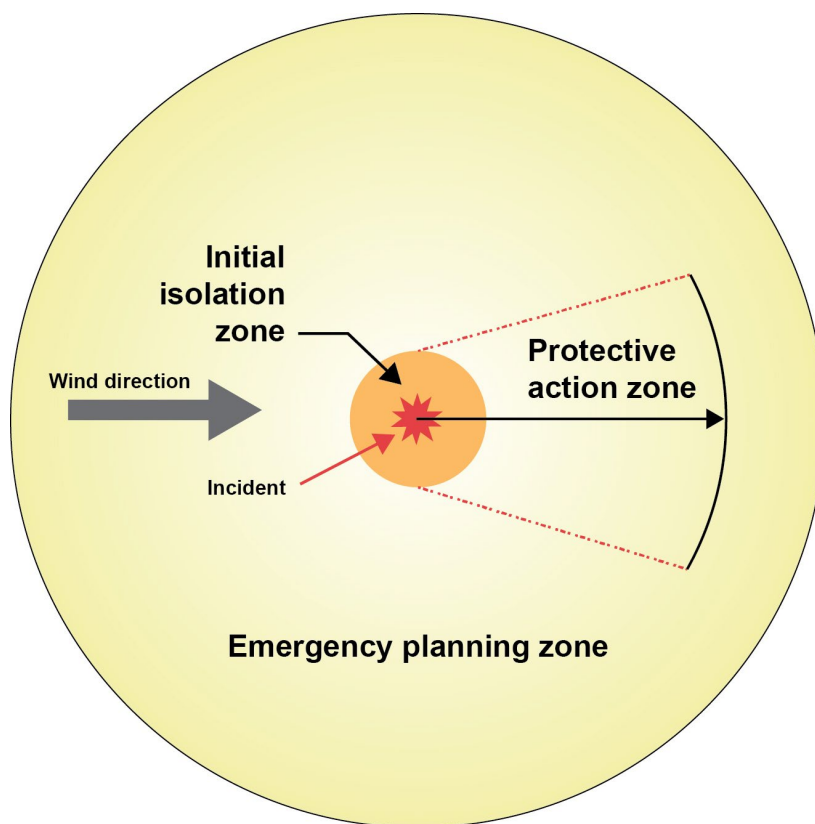


Figure 1. Example IIZ, PAZ, and EPZ

10 H₂S and HVP Products – Emergency Preparedness Involving the Public and Local Authority

This section sets out the requirements for emergency preparedness involving the public and local authorities for approval holders whose operations contain H₂S or HVP products or both. Table 1 lists the situations for involving the public and local authorities, including notification and engagement.

Engagement processes use methods of communicating with the public that are acceptable to the AER; such engagement should enable participants to confirm their understanding of the discussions. Examples of acceptable forms of engagement include face-to-face meetings (i.e., in-person engagement), email, video conferencing, registered or regular mail, telephone, and open houses.

Conduct in-person engagement with the public who wish to participate in the ERP development process. Future engagement expectations can be determined during these meetings.

Notification during the public preparedness program involves providing the recipient with a copy of the public emergency preparedness information package.

Table 1. When to notify and engage the public or local authorities

Situation	Notification and engagement actions
Developing an operation-specific ERP requiring AER approval	Notify and engage in person with the public within the EPZ before submitting an application to the AER for approval. Engage with the local authority to confirm and coordinate each party's roles and responsibilities.
Change in the overall EPZ size ^a	EPZ decrease: Notify the public and local authorities who are no longer within the overall EPZ. EPZ increase: Notify and engage in person with the public and local authorities who are in the expanded portion of the overall EPZ.
Drilling the first sour zone or nonconsecutive sour well completion	Notify the public within the EPZ at least 24 hours before operations begin.
Delayed drilling, completion, and servicing operations	Notify and conduct engagement processes with the public and local authority if a well ERP is to remain active beyond one year from approval.
Workovers with wellhead removal	Notify the public who wanted to leave before the wellhead-off workovers at least 48 hours before operations begin.
End of an operation-specific ERP for sour drilling/completion/servicing operations	Notify the public and local authorities included in the ERP. Archive the ERP in the designated information submission system.
Modifications to an existing facility	Notify and engage with the public within the EPZ and local authority if modifications to existing facilities result in a size change to the EPZ or changes to facility response procedures.
Temporary surface pipeline	Notify and engage with the public within the temporary surface pipeline EPZ.
Transfer of ownership	Notify the public and local authorities within the EPZ after finalizing a sale but before transferring wells, facilities, or pipelines requiring an operation-specific ERP. Notify and engage in person with the public within the EPZ before applying for AER approval. Engage the local authority to coordinate roles and responsibilities.
Cancelling an operation-specific ERP	Notify all public and local authorities included in the operation-specific ERP. Archive the operation-specific ERP in the designated information submission system.
Plan management: existing operation-specific ERPs	Conduct engagement processes with the public and local authorities every two years for operation-specific ERPs for sour operations, HVP pipeline , or cavern storage ERPs.

^a An EPZ is set for a specific well, pipeline, or facility. An overall EPZ covers multiple wells, pipelines, or facilities included in one operation-specific ERP.

10.1 Emergency Preparedness Involving the Public

79) The approval holder must notify or engage with the public due to changes specified in table 1.

10.1.1 Public Emergency Preparedness Information Package

80) Where an operation-specific ERP is required, the approval holder must develop a public emergency preparedness information package and include it as part of the operation-specific ERP application submission to the AER (section 11). The package must include the following:

- a) an EPZ map of the operations in the area
- b) range of release rates, release volumes, H₂S concentrations (if applicable), and EPZ determinations for their wells, pipelines, and facilities
- c) a description of potential health effects that could result from exposure to H₂S (see appendix 2), HVP product, or **sulphur dioxide** (SO₂) if applicable (refer to documents such as the Alberta Health Services publication, [*Alberta Health Acute Exposure Health Effects of Hydrogen Sulphide and Sulphur Dioxide*](#))
- d) public protection measures that may be implemented (includes information on special emergency procedures unique to the community, such as those used by hospitals and schools, which could affect emergency response)
- e) contact information for a representative on behalf of the approval holder
- f) information for managing potential ignition sources during an HVP release (if applicable)

10.1.2 Public Preparedness Program

10.1.2.1 Notification

81) The approval holder must notify the public identified in table 2 that they are included in the approval holder's operation-specific ERP and provide them with a copy of the public emergency preparedness information package.

Table 2. Who to notify and engage in the public preparedness program

Notify and Engage	Notify Only	Notify Only if Requested
- Permanent and part-time residents, including those residing on dead-end roads outside the EPZ but egress through the EPZ. - Private and public recreational property owners, operators, and occupants within the EPZ. - Business owners and operators within the EPZ, including AER-regulated operators with staffed facilities. - Operators of rural public facilities and publicly used developments within the EPZ. - Local authorities within the EPZ.	Registered trappers, registered grazing lease and allotment users, and nonresident owners of rented residences whose properties are within the EPZ.	- Members of the public inside or close to the EPZ who have expressed to the approval holder a desire to receive notification, including nonresident landowners, farmers renting land, recreational users, guides, and outfitters. - Industrial operators without manned facilities in the EPZ area who have expressed to the approval holder a desire to receive notification. - Health authority or first responder agency who have expressed to the approval holder a desire to receive notification.

10.1.2.2 Engagement

- 82) When developing an operation-specific ERP for AER approval or when a new **surface development** is identified or added to an existing operation-specific ERP, the approval holder must take the following actions:
- Attempt to conduct in-person engagement with all individuals specified in table 2.
 - Review key incident response information with the entities identified within the EPZ who wish to participate in the engagement process to familiarize them with potential emergencies and corresponding public protection measures for emergency response procedures. The approval holder's representative is expected to have the necessary knowledge to provide details of the emergency response procedures and address any questions and concerns.
 - Offer to engage with residents who do not wish to meet in person by alternative means of communication.
 - Provide residents with a public emergency preparedness information package using a trackable method (e.g., registered mail) unless they prefer regular mail or choose not to participate in the engagement process.
 - Address any request for additional information or modifications of the ERP by the individual engaged.

- 83) If the approval holder cannot contact the parties noted in table 2, or a party is unwilling to provide emergency contact information, the approval holder must account for them in the ERP and demonstrate they made reasonable efforts to meet with all parties.
- 84) If engagement necessitates changing an operation-specific ERP, the approval holder must inform and discuss the changes with the local authority.
- 85) The approval holder must notify residents of urban centres that they are within the EPZ and provide details of the public protection measures available in an emergency.

Approval holders may use a combination of methods to notify residents in urban areas, such as mailouts, open houses, or other communication technology, based on the preference of the local authorities to notify residents in urban areas. The approval holder is encouraged to coordinate notification with the municipality.

- 86) For an existing operation-specific ERP for **sour gas** operations, HVP pipelines, or cavern storage facilities, the approval holder must conduct a public preparedness program as outlined in table 2 every 24 months from the date of ERP approval through the engagement process detailed in table 1.
- 87) For existing operation-specific ERPs for sour gas drilling, completion, or servicing operations, the approval holder must conduct a public preparedness program as outlined in table 2 every 12 months from the date of the operation-specific ERP approval through the engagement process detailed in table 1.
- 88) Regardless of whether residents wish to meet with the approval holder, the approval holder must attempt to provide an up-to-date public emergency preparedness information package to each residence. The approval holder must
 - a) provide emergency response information,
 - b) review public protection measures, and
 - c) answer any concerns or questions from residents.

10.1.3 Incorporating Information into an Operation-Specific ERP

- 89) The approval holder must attempt to obtain the following information through the public preparedness program for inclusion in the operation-specific ERP requiring AER approval:
 - a) the exact location of the residence, place of business, or **public facility**, including egress route issues (legal description or address)
 - b) name of key contact and a 24-hour contact telephone number (home, business, cell phone, or other), and alternate contact if possible

- c) number of residents
 - d) names of those that may require response assistance—persons can be considered to require response assistance and require early notification or **evacuation** without having to divulge personal health issues
 - e) any other information deemed necessary to allow for effective emergency response procedures to be developed
- 90) The approval holder must consider any resident unwilling to provide personal information as requiring response assistance.

The approval holder's representative is expected to inform the public that they can seek evacuation assistance without divulging personal health issues and receive early notification or evacuation.

The approval holder may collect personal information solely to maintain public safety during an incident and provide this information to emergency responders and the AER. Personal information in the approval holder's possession is governed by the *Personal Information Protection Act*. Once filed with the AER, it is subject to the *Protection of Privacy Act*.

See section 9.2 of *Manual 026* for guidance on the public preparedness program.

10.2 Emergency Preparedness Involving a Local Authority

10.2.1 Local Authority Emergency Preparedness Information Package

The approval holder is encouraged to establish a working relationship with the local authorities to coordinate emergency response resources and enhance emergency response efficiency. See section 9.3 of *Manual 026* for details.

- 91) Where an operation-specific ERP is required, the approval holder must develop a local authority emergency preparedness information package. The package must include the following:
- a) an overview of the approval holder's public preparedness program
 - b) the roles and responsibilities of both parties during an incident
 - c) when a local authority wishes to be contacted for planning activities
 - d) the preferred method of contact for the local authority for preparedness and response
 - e) limitations or capabilities of both parties for incident response
 - f) requests for changes to the approval holder's operation-specific ERP from the local authority
 - g) a summary of decisions made

- h) a summary and action plan to resolve any disagreements between the approval holder and the local authority
- 92) The approval holder must provide local authorities with a copy of the local authority emergency preparedness information package unless requested otherwise.

10.2.2 Engaging Local Authorities

- 93) The approval holder must engage with the local authorities at least once every 24 months to coordinate emergency response resources and enhance emergency response efficiency unless requested otherwise.
- 94) When a local authority agrees to engage, the approval holder must review the components of its preparedness information package with the local authority, document any decisions, and incorporate response-related procedures requested by the local authority into its operation-specific ERP.
- 95) As part of its engagement with the local authority, the approval holder must discuss how notification and evacuation will occur outside the EPZ and include a summary of the engagement in the operation-specific ERP.

If an EPZ includes a portion of an urban centre, the approval holder is not required to identify each residence within the urban centre; however, contact must be made with the appropriate urban directors of emergency management to review key emergency response information and confirm and coordinate each party's roles and responsibilities.

Approval holders are encouraged to streamline emergency preparedness activities with local authorities. For example, where local authorities are associated with multiple operation-specific ERPs, conduct a single engagement with the local authorities instead of multiple engagements.

10.3 Record Keeping

- 96) As part of the emergency preparedness with local authorities and the public preparedness program, the approval holder must have a process for recording the following:
 - a) discussions with the appropriate authorities
 - b) the method of notification to residents in urban centres, including a sample of the notice
 - c) the number of attempts made and the method used to contact an individual or local authority if the approval holder was unable to make contact
 - d) record of engagement with the public, including unsuccessful attempts to engage or obtain the cooperation of any required persons and any outstanding issues yet to be resolved, and the date of contact

See section 11.5 for the requirements for keeping operation-specific ERPs up to date.

11 H₂S and HVP Products – Requirements for Operation-Specific Emergency Response Plans

This section sets out requirements for an operation-specific ERP for approval holders whose operations involve H₂S and HVP products.

11.1 Operation-Specific ERP Submission

- 97) The approval holder must submit an operation-specific ERP and the “ERP Application” form (see the directive [landing page](#)) to the AER via the designated information submission system for approval if any of the following conditions apply:
 - a) The EPZ includes surface developments or egress routes through the EPZ for residents of nearby residences.
 - b) The operation includes critical sour wells.
 - c) The operation includes a cavern storage facility storing HVP products.
- 98) For sour wells, until a hard copy or electronic copy of all applicable operation-specific ERPs and the AER approval letter is available on site for the duration of the operations, the approval holder must not
 - a) spud the well, drill out the surface casing, or remove any component of an existing wellhead;
 - b) conduct testing; or
 - c) perform workovers, completions, or well servicing.
- 99) Where an operation-specific ERP is required, the approval holder must ensure it is approved before starting operations.

Manual 026 provides examples of when an ERP requires AER approval.

11.2 Supplements to an Existing Operation-Specific ERP

When additional operations or assets are constructed or acquired, the approval holder may submit a supplement to an existing operation-specific ERP instead of submitting a new operation-specific ERP. However, supplements are temporary and incorporated into the operation-specific ERP in accordance with requirement 19.

Examples of when supplements may be used include the following:

- new well or pipeline tie-ins or facilities

- new operating areas
- significant changes to the EPZ or new residences within the EPZ
- drilling or completion operations
- sour well workovers, well servicing, and testing

The examples in *Manual 026* of situations where an operation-specific ERP requires AER approval also apply to ERP supplements requiring AER approval.

- 100) Where a supplement to an operation-specific ERP requires AER approval, the approval holder must submit the supplement to the AER for approval via the designated information submission system before starting operations unless the added wells or operations are currently operating.

An ERP supplement is not required for reapproval if it is for new operations added to an existing operation-specific ERP where the EPZ size remains unchanged or no surface developments are present within the expanded portion of the EPZ.

- 101) Where a supplement to an operation-specific ERP requires AER approval, the approval holder must
- a) conduct a public and authority preparedness program for all surface developments within the proposed well's EPZ when the supplement is for drilling and completions for sour wells, and
 - b) for all other operations, conduct a public preparedness program for all new surface developments in the portion of the EPZ extending beyond the existing EPZ boundary.

11.3 Multiwell Programs

An operation-specific ERP may be developed and submitted for multiwell programs if they include site-specific information for each well. The approval holder may develop common procedures and response infrastructure for the entire program, provided the operation-specific ERP remains current throughout the project and is maintained in accordance with this directive.

11.4 Temporary Surface Pipelines

An approval holder may include a temporary surface pipeline in its operation-specific ERP for sour well site-specific drilling or completion, provided the pipeline EPZ is within the well's EPZ.

- 102) The approval holder must develop and submit a separate operation-specific ERP for approval where the temporary surface pipeline EPZ extends beyond the well EPZ boundary and includes surface developments.

11.5 Managing an Operation-Specific Emergency Response Plan

The purpose of the management process is to ensure operation-specific ERPs are kept up to date and provided to the AER, the appropriate authorities, and the public, as defined in this directive, who have requested a copy.

The AER does not approve annual operation-specific ERP updates but audits approval holders to verify the accuracy of their operation-specific ERPs. Out-of-date operation-specific ERPs that could result in ineffective emergency response will be subject to enforcement action.

- 103) In addition to requirement 19, the approval holder must demonstrate that its plan management process will keep its operation-specific ERPs up to date and must include the following activities:
 - a) Conduct ground truthing every 12 months to verify changes, such as new residents, businesses, and renters, and verify the operation-specific ERP maps. The approval holder may use any method for ground truthing.
 - b) Contact residents within the EPZ to update their information.
- 104) In accordance with requirement 20, the approval holder must notify those persons holding copies of an operation-specific ERP of the completion of an operation and that the operation-specific ERP is no longer in effect. The notification includes the AER and all residents within the EPZ who requested notification of completion.
- 105) The approval holder must engage the parties noted in table 2 no later than every 24 months as part of its maintenance of an operation-specific ERP.

Refer to section 9.4 of *Manual 026* for examples of collaborative approaches to ERP management.

11.6 Distribution of Operation-Specific ERPs

Once the AER approves an operation-specific ERP, including approval of supplements, it becomes a public document subject to disclosure, except for confidential resident and personal information.

- 106) The approval holder must provide a copy of the operation-specific ERP, excluding confidential resident and personal information, within ten business days of receiving AER approval to
 - a) any resident within the EPZ who requests in writing to have a copy,
 - b) the appropriate authorities who request in writing to have a copy, and
 - c) primary approval holder response personnel.

11.7 Critical Sour Well Review Meetings for Operation-Specific ERPs

- 107) For all critical sour wells and zones, the approval holder must conduct a review meeting before the commencement of operations to
- review the hazards associated with the operation (from the hazard identification done previously) and any potential new hazards,
 - review the roles and responsibilities of response personnel, and
 - assess on-site personnel capabilities required to implement the ERP.
- 108) The approval holder must notify the AER via the designated information submission system at least 96 hours before the intended review meeting date and local authorities at least five business days before the review meeting date.
- 109) For all critical sour well drilling and completion, workover, and well servicing operations, the approval holder must conduct a meeting within 96 hours before entering the first sour zone to identify hazards associated with the operation, review roles and responsibilities, and assess on-site personnel capabilities required to implement the ERP. Required personnel include
- field response personnel with assigned roles and responsibilities in the ERP and
 - key personnel involved in supervising and managing incident response.
- 110) If drilling, completion, workover, or well servicing operations include a critical sour zone or a combination of zones that makes the well a critical sour well, the approval holder must conduct a meeting concerning the critical sour zone work before entering the sour zone.
- 111) The approval holder must notify the AER via the designated information submission system at least four business days before the meeting for critical sour zone work and at least four business days' notice of the meeting to the local authority.

All meetings concerning critical sour well drilling, completion, workover, and well servicing operations should include the following:

- verification of the assigned roles and responsibilities in the ERP
- identification of any revision to the ERP
- confirmation that the emergency contact numbers are correct
- communication of EPZ size and EPZ inventory to well-site personnel

See section 10.3 of *Manual 026* regarding review meetings and sour operations.

12 H₂S and HVP Products – Content for an Operation-Specific Emergency Response Plan

This section sets out requirements for the content of an operation-specific ERP for approval holders whose operations involve H₂S and HVP products.

In addition to the information required in section 5.3, an operation-specific ERP includes the following content:

- an ERP overview
- operational information
- maps
- incident detection, notification, and confirmation procedures
- hazard monitoring procedures during an incident
- isolation procedures
- procedures for **sheltering in place** or evacuation
- ignition procedures (if required)

12.1 ERP Overview

112) The approval holder's operation-specific ERP must include the following information:

- a) an introduction and summary
- b) a brief overview of the operations
- c) the name of the operating area or system
- d) a schedule (e.g., expected spud date, date of entry into the sour zone, and estimated time in the sour zone), if relevant to the operation-specific ERP
- e) general land use in the surrounding area, including population density, number of residents, level of transient usage, public facilities, and roads
- f) the distance to the nearest urban centre and residence

12.2 Operational Information

113) The approval holder's operation-specific ERP must include the following information about its operations:

- a) name and legal descriptions of the locations and type of operation
- b) licence or approval numbers for wells, facilities, and pipelines, and pipeline line numbers

- c) operational status
- d) identified hazards with the potential to result in high consequences to public safety and the environment
- e) size and configuration of the EPZ
- f) if applicable, sour wells (including critical sour wells), **sour pipelines**, and facilities information, including
 - i) surface location of wells, emergency shutdown (ESD) valves, pipelines, and facilities
 - ii) name and legal description of wells (bottomhole location), pipelines, and facilities
 - iii) maximum cumulative H₂S release rates, volumes, concentrations, and level designation per *Directive 056*
- g) if applicable, HVP cavern storage facility and HVP pipeline information, including
 - i) name or legal description of facilities and pipelines
 - ii) maximum cumulative HVP release volume or release rate for the facility
 - iii) routing maps of the pipelines
 - iv) maximum potential HVP release volumes for all pipeline segments within the system
 - v) maximum licensed operating pressure, internal diameter, and minimum operating temperature of the pipelines
 - vi) length of pipeline between ESD valves

12.3 Maps

- 114) The approval holder's operation-specific ERP must include maps scaled to represent the entire mapped area with the following information:
- a) surface locations of the operations and access roads
 - b) the EPZ boundary
 - c) resident reference numbers for residences within the EPZ, residences within 25 m outward of the EPZ boundary, and residences on dead-end roads where residents would need to egress through the EPZ
 - d) provincial, local, and access roadways and dead ends in the EPZ
 - e) lakes, rivers, streams, and wetlands
 - f) **urban density developments**, campgrounds, recreation areas, public facilities (e.g., churches, schools, hospitals), and any other publicly used development within the EPZ

- g) trapping areas, grazing leases, and range allotment boundaries and their reference numbers
- h) other industrial operations, including oil and gas operations
- i) railways and airports
- j) municipal boundaries (e.g., hamlets, villages, towns)
- k) health authority boundaries
- l) a legend, scale, and north directional indicator
- m) potential roadblocks for sour well site-specific drilling and completions

12.4 Incident Notification

See section 8 for requirements related to initial incident procedures.

- 115) The communications plan (see section 5.3.3) in the approval holder's operation-specific ERP must include specific procedures for how and when notification will occur within the EPZ.

12.5 Air Quality Monitoring Procedures During an Incident

- 116) The approval holder's operation-specific ERP must
 - a) provide details on the intended use and procedures to activate air quality monitoring equipment, such as stationary and **mobile air quality monitoring units** and personal handheld monitors, and
 - b) procedures for continuously monitoring the dispersion of hazardous substances during an incident.
- 117) The approval holder's operation-specific ERP must specify the type and number of hazard monitoring devices appropriate for use during an incident to address the following considerations:
 - a) hazard type
 - b) entry and egress points
 - c) population density and distance to urban centres or rural subdivisions
 - d) local weather conditions and topographical features

12.6 Access Control Procedures

- 118) The approval holder's operation-specific ERP must include procedures to restrict unauthorized and unprotected entry into the emergency response zones during an incident.

- 119) The approval holder's operation-specific ERP must identify any special procedures needed to address major highways and railways passing through the EPZ that could be affected by a hazard.

12.7 Shelter-in-Place or Evacuation Procedures

- 120) The approval holder's operation-specific ERP must address how sheltering in place within or evacuation from a response zone will be accomplished during an incident, including how land users (traditional or otherwise), such as hunters, trappers, recreational users, and nonresident landowners, will be found, instructed, and evacuated.
- 121) The approval holder's operation-specific ERP must address any special procedures required for evacuating public facilities involving large numbers of people, including assistance with transportation or changes in the normal notification procedures.
- 122) The approval holder's operation-specific ERP must include shelter-in-place instructions.

12.8 Ignition Procedures

- 123) The approval holder's operation-specific ERP must include the following information concerning the ignition of sour gas wells and HVP product releases:
 - a) The ignition decision procedures (e.g., ignition criteria flowchart), including a description of the equipment to be used in the event ignition criteria are met.
 - b) An acknowledgement that one of its on-site representatives has been designated as the ignition authority.
- 124) In addition to requirement 123, the approval holder's operation-specific ERP must address the following concerning releases from sour gas wells and HVP product releases from a pipeline or cavern storage facility:
 - a) The approach to identifying the location of a hazardous plume.
 - b) The factors to consider when deciding whether to ignite a release, such as changing weather conditions.
 - c) The ignition decision procedures, including a copy of figure 2.
 - d) Protocols to support a decision to ignite a release, including emergency response procedures for immediate ignition and actions taken if a release occurs while personnel are on site.
 - e) The actions required before attempting to ignite a dispersing HVP product plume, such as establishing the boundary of the dispersing vapour cloud.

- f) The names or position titles of personnel authorized to conduct the ignition, and a description of the follow-up actions.

13 H₂S and HVP Products – Incident Management

This section sets out incident management requirements for approval holders whose operations involve H₂S and HVP products.

- 125) The approval holder must have up-to-date copies of the applicable corporate ERP and any operation-specific ERPs (hard copy or electronic) available at a response location.
- 126) The approval holder must assign roles and responsibilities to personnel under its corporate ERP and any operation-specific ERPs to ensure their proper implementation during an incident.
- 127) The approval holder must follow and implement the procedures and response actions outlined in its corporate ERPs and any operation-specific ERPs.

13.1 Initial Response

- 128) Where sour operations are conducted at two or more sour wells with overlapping EPZs and surface developments are present within the EPZs, if an approval holder declares an emergency, the other approval holders must also suspend operations until it is over.

13.2 Communication During an Incident

- 129) The approval holder must make the information in appendix 4 available to the public as soon as possible during an incident.

13.3 Air Quality Monitoring

- 130) During drilling, completion, testing, or servicing operations in potentially critical sour zones where the EPZ includes a portion of an urban density development or urban centre, the approval holder must ensure the following:
 - a) One air quality monitoring unit is nearby.
 - b) Dispatch another monitoring unit when loss of well control is evident and sour gas release is likely.
 - c) One unit monitors the boundary of the urban density development or urban centre, and the other tracks the plume.

- 131) During drilling, completion, testing, or servicing operations for all noncritical sour wells and for operations in potentially critical sour zones where the EPZ does not include a portion of an urban density development or urban centre, the approval holder must dispatch an air quality monitoring unit when loss of well control is evident and sour gas release is likely.

See section 11.1 of *Manual 026* regarding AER expectations for air quality monitoring during an incident.

13.4 Public Protection Measures

- 132) The approval holder must initiate the public protection measures shown in figure 2 for incidents originating from its operations that may affect the public.

The type of public protection measures used depends on the severity of the incident and the monitored results in unevacuated areas.

13.4.1 Notification During an Incident

- 133) If a sour gas or HVP product release has the potential to affect an area beyond the lease, facility boundary, or pipeline right-of-way, the approval holder must notify
- a) the public in the portion of the emergency response area that is within the EPZ,
 - b) the local authority's director of emergency management, if an urban centre is within the EPZ,
 - c) individuals within the EPZ who have requested early notification and wish to evacuate voluntarily, and
 - d) the appropriate authorities.

The approval holder's personnel or designated responders may be required to provide evacuation assistance to individuals as requested.

- 134) Once notification attempts are complete in accordance with requirement 133, the approval holder must notify the public in the remainder of the EPZ (i.e., any person not previously notified).
- 135) The approval holder must advise the public within the EPZ of the public protection measures required.

The approval holder may use the notification mechanisms in the appropriate authority's emergency plan and program if agreed to by the appropriate authority.

13.4.2 Shelter in Place or Evacuation

136) The approval holder must assess the following before advising a resident to shelter in place or evacuate:

- a) the size and expected duration of the release
- b) egress routes
- c) current and expected meteorological conditions
- d) the potential for unexpected ignition

Figure 2 is a decision tree of the public protection measures.

13.4.2.1 Sour Gas Releases

137) The approval holder must continuously assess and act on the need to expand the evacuation area based on the monitored levels of H₂S and SO₂ and take the actions noted in table 3.

Table 3. Approval holder actions for H₂S and SO₂ concentrations

Concentrations in unevacuated areas	Actions in the unevacuated areas
H ₂ S of 1 to 10 ppm (3-minute average)	Notify persons who requested notification so they may voluntarily evacuate before exposure to H ₂ S.
H ₂ S above 10 ppm (3-minute average)*	Assess local conditions and notify all persons to evacuate or shelter in place.
SO ₂ of:	Notify all persons to evacuate immediately.
• 5 ppm (15-minute average) • 1 ppm (3-hour average) • 0.3 ppm (24-hour average)	

* If monitored levels over the 3-minute interval are declining (i.e., three readings show a decline from 15 ppm to 10 ppm to 8 ppm over 3 minutes), evacuation may not be necessary even though the average over the 3-minute interval would be 11 ppm. Approval holders should use proper judgement in determining if evacuation is required.

138) In the absence of monitored readings, responders must advise residents to shelter in place.

When safe to do so, evacuation should take place before a release of sour gas or HVP product has the potential to affect people in proximity to the release or as soon as possible to avoid any exposure to the hazard.

If evacuation is not possible, sheltering in place can protect the public under certain conditions.

139) The approval holder must advise residents to evacuate if required.

The approval holder may only advise residents to evacuate; however, the local authority must declare a **state of local emergency** before mandatory evacuation can occur.

13.4.2.2 HVP Product Releases

Sheltering in place is the primary public protection measure for an HVP product release outside the IIZ. Evacuating the public should only proceed when it is safe to do so and after an assessment in accordance with figure 2.

13.4.3 Notification and Evacuation Outside the EPZ

140) If public protection measures are required beyond the EPZ, the approval holder must conduct them in accordance with the arrangement with the local authority.

The Government of Alberta may activate the [*Energy Resources Industry Emergency Support Plan*](#) for level 2 and level 3 emergencies to support incident response. The notification mechanisms will be based on monitored air quality and other situations that might arise during the incident.

Evacuation of the area outside the EPZ is coordinated through the approval holder's operation-specific ERP and the response framework in the local authority's emergency plans and programs. The health authorities also have a role in evacuation in accordance with section 52.2 of the [*Public Health Act*](#).

13.4.4 Activating the Registration and Support Centre

141) Unless directed by local authorities otherwise, the approval holder must

- a) activate a registration and support centre situated a safe distance from the release source and
- b) meet and register evacuees at the registration and support centre.

13.4.5 Ignition

142) The approval holder must ensure the ignition team is trained and competent in sour gas or HVP product ignition and has the proper equipment to ignite the release within the time limits for which the EPZ was designed.

Ignition criteria are provided in appendix 5, and the decision process is shown previously in figure 2. The AER may direct the approval holder to ignite a release if the approval holder does not agree to ignite the release or is not prepared to take the necessary steps.

13.4.5.1 Sour Well Release Ignition

143) The approval holder must ensure

- a) all sour wells have an ignition system, such as a flare gun, on site during drilling, completion, well testing, or workover operations in sour zones and

- b) all critical sour wells have a dual-ignition system on site during drilling operations in critical sour zones and during completion, well testing, or workover operations when the wellhead is removed.
- 144) The approval holder must
- a) keep the AER representatives informed about the ignition situation and ignite a sour gas flow to the atmosphere in accordance with the ignition assessment criteria in appendix 6, unless the AER determines that ignition may be delayed, and
 - b) assign the decision-making authority to an on-site representative to ignite the release.

The primary ignition system installed at a critical sour well should enable remote activation from a safe location through a triggering device. The secondary system may be a manual system, such as a flare gun.

Ignition does not negate the need to continue evacuation, as residual pockets of H₂S or SO₂ may be in the area.

13.4.5.2 Pipeline or Cavern Storage Facility HVP Product Releases and Ignition

Following an incident, the hazard associated with an HVP product release may be controlled or minimized by deliberately igniting the release.

- 145) When deciding to ignite HVP releases, the approval holder must consider the following:
- a) the increased risks of a delayed ignition
 - b) whether the perimeter of the hazard area has been established
 - c) whether the public has been evacuated from the area
 - d) whether ignition will worsen the situation by endangering the public or the environment, or damage equipment used to control the product
 - e) whether wind direction has been established and continuously monitored
 - f) whether the possibility of an explosion has been assessed (i.e., obstructions or regions of congestion within the perimeter of the dispersing vapour cloud)
- 146) The approval holder must assign the decision-making authority to an on-site representative to ignite an HVP product release.

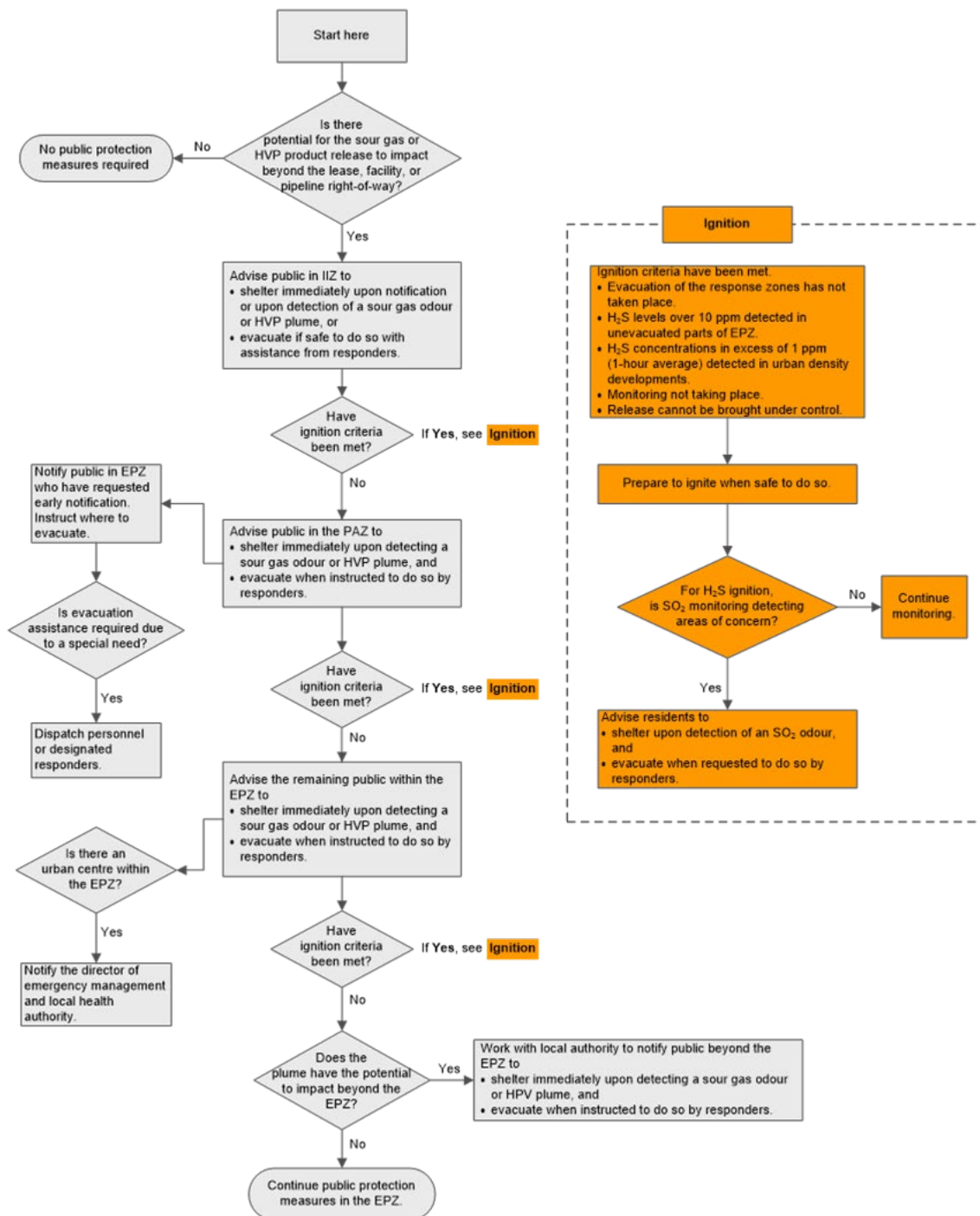


Figure 2. Public protection measures for H₂S and HVP products

Appendix 1 Definitions

alert	See appendix 3.
appropriate authority	The authority(ies) with jurisdiction, including local authorities as defined in the Emergency Management Act , health authorities, governments and agencies, police, and others with a role in providing effective incident response.
approval holder	A licensee as defined in Oil and Gas Conservation Act (OGCA), Oil Sands Conservation Act (OSCA), Coal Conservation Act , Pipeline Act , Geothermal Resource Development Act (GRDA), and Mineral Resource Development Act (MRDA); an approval holder as defined in the OGCA, CCA, and MRDA; a permittee as defined in the CCA and MRDA; an operator as defined in OSCA.
cavern	A cavern used for hydrocarbon storage.
corporate ERP	The plans prepared by an approval holder under the OGCA, OSCA, Pipeline Act, GRDA, MRDA, or any other regulation, statute, or condition imposed by the AER. A corporate ERP is used when planned procedures allow for effective incident response, and an operation-specific ERP is not required.
critical sour well	A well with an H ₂ S release rate greater than two cubic metres per second or wells with lower H ₂ S release rates near an urban centre as defined in Directive 056: Energy Development Applications and Schedules .
emergency	As defined in the Oil and Gas Conservation Rules , Oil Sands Conservation Rules , Pipeline Rules , Brine-Hosted Mineral Resource Development Rules , Rock-Hosted Mineral Resource Development Rules , and Geothermal Resource Development Rules .
emergency management program (EMP)	As defined in the <i>Oil and Gas Conservation Rules</i> , <i>Oil Sands Conservation Rules</i> , <i>Pipeline Rules</i> , <i>Brine-Hosted Mineral Resource Development Rules</i> , <i>Rock-Hosted Mineral Resource Development Rules</i> , and <i>Geothermal Resource Development Rules</i> .
emergency planning zone (EPZ)	An EPZ is a geographic area around wells, pipelines, or facilities where the presence of hazardous substances requires specific emergency preparedness by the approval holder.
emergency response plan (ERP)	A comprehensive plan to protect the public that includes criteria for assessing an incident (emergency) and procedures for mobilizing response personnel and agencies and establishing communication and coordination among the parties.
emergency response zones	The IIZ and PAZ collectively.

engagement processes	Communication methods where the participants understand the discussions that took place.
evacuation	An organized, phased, and supervised withdrawal of persons from dangerous or potentially dangerous areas to safe areas.
facility	As defined in the Oil and Gas Conservation Act , Oil Sands Conservation Act , Coal Conservation Act , Geothermal Resource Development Act , and Mineral Resource Development Act .
functional exercise	An activity designed to evaluate capabilities and multiple functions using simulated response.
hazard	A potential source of harm to people, the environment, or property.
high-vapour-pressure (HVP) pipeline	A pipeline system to convey hydrocarbons or hydrocarbon mixtures in a liquid or quasi-liquid state with a vapour pressure greater than 110 kilopascals absolute at 38°C using the Reid method. See ASTM D 323, <i>Standard Test Method for Vapor Pressure of Petroleum Products (Reid Method)</i> .
hydrogen sulphide (H₂S)	A naturally occurring gas found in geological formations or formed by the natural decomposition of organic matter in the absence of oxygen. It is colourless, has a molecular weight heavier than air and is highly toxic. In small concentrations, it smells like rotten eggs and causes eye and throat irritation. Depending on the particular mix of gases, gas properties, and ambient conditions, a sour gas release may • be heavier than air (dense), so it will tend to drop towards the ground with time, • be lighter than air (buoyant), so it will tend to rise with time, or • be about the same weight as air (neutrally buoyant), so it will tend to neither rise nor drop but will disperse with time.
incident	A situation that might be, or could lead to, a disruption, loss, emergency, or crisis.
incident classification	A system that examines the risk level to the public of an incident and assigns an incident level based on the consequence of the incident and the likelihood of escalation.
incident command system (ICS)	A system used to coordinate preparedness and incident management.
initial isolation zone (IIZ)	The area in proximity to a hazardous release where indoor sheltering may provide temporary protection due to the proximity of the release.
level 1 emergency	See appendix 3.
level 2 emergency	See appendix 3.
level 3 emergency	See appendix 3.
local authority	As defined in the Emergency Management Act .

major exercise	An exercise involving approval holders and emergency response agencies involving the deployment of response resources to test the approval holder's ERP. The exercise intends to provide a realistic simulation of incident response.
mobile air quality monitoring unit	Portable equipment to measure hazardous substances (e.g., H ₂ S or SO ₂) at very low atmospheric concentrations of parts per billion.
mutual aid understanding	An understanding between two or more public or private parties (e.g., oil and gas companies, service companies, and local authorities) that defines each party's commitment to providing aid and support during an incident.
operation-specific ERP	As defined in the <i>Oil and Gas Conservation Rules, Oil Sands Conservation Rules, Pipeline Rules, Coal Conservation Rules, Brine-Hosted Mineral Resource Development Rules, Rock-Hosted Mineral Resource Development Rules, and Geothermal Resource Development Rules</i> .
preparedness	A state of readiness to respond to an emergency. This is achieved through actions taken before an emergency with a goal of an effective response.
program coordinator	The person appointed by the approval holder to develop, implement, evaluate, and maintain the EMP and all ERPs for the approval holder.
protective action zone (PAZ)	An area downwind of a hazardous release where outdoor pollutant concentrations may result in life-threatening or serious irreversible health effects on people.
public	The people who are or may experience the effects of an emergency. Public is an umbrella term. The composition of the public in the area affected by an emergency from an approval holder's operations may differ from area to area and may include stakeholders or Indigenous groups.
public facility	A building, such as a hospital, rural school, or major recreational facility, situated outside of an urban centre that can accommodate more than 50 individuals or where additional transportation may be required during an evacuation.
public protection measures	Sheltering, evacuation, ignition, and isolation procedures to mitigate the effects of a hazardous release on the public.
publicly used development	Places where 50 individuals or less may be expected (e.g., businesses, cottages, campgrounds, churches, Indigenous cultural or ceremonial sites, and other locations created for use by the nonresident public).
residence/resident	An occupied dwelling/persons in the residence. These terms are used in relation to occupants and dwellings within an EPZ.
response	Actions immediately before, during, or after an emergency to manage the consequences.

senior management	Persons (e.g., chief executive officer, president, board member, or director) with authority to influence the direction and culture of the organization.
sheltering in place	To remain indoors for short-term protection from exposure to toxic gas releases.
spill cooperative	A member-funded organization that providing spill response support services meeting the requirements of this directive.
sour gas	Natural gas, including solution gas, with H ₂ S.
sour pipeline	A pipeline to convey gas or liquid, or both, that contains sour gas.
sour well	An energy resource or mineral resource well where formations bearing sour gas may be encountered during drilling or an oil or gas well capable of producing sour gas.
state of local emergency	A declaration by a local authority under the Emergency Management Act or by the medical officer of health under the Public Health Act to enable resources and procedures at the municipal level to effectively and efficiently resolve an emergency.
substance	As defined in Environmental Protection and Enhancement Act .
sulphur dioxide (SO₂)	A colourless, water-soluble, suffocating gas formed by burning sulphur in air and is also used to manufacture sulphuric acid. SO ₂ has a pungent smell, like a burning match. SO ₂ is extremely toxic at higher concentrations. The molecular weight of SO ₂ is heavier than air; however, typical releases are related to combustion, which makes the gaseous mixture lighter than air (buoyant).
surface development	Residences that are occupied full time or part-time, publicly used developments, public facilities (including registered campgrounds and places of business), and any other surface development where the public may gather regularly. Surface development includes residences within 25 m outward of the EPZ boundary and those from which residents are required to egress through the EPZ.
synergy group	Community-based, multistakeholder groups that work collaboratively in addressing the pressures of resource development in ways that support the social, environmental, and economic well-being of their communities and the province.
tabletop (exercise)	An exercise to review resource allocations and the roles and responsibilities of response personnel. This exercise enables new personnel to familiarize themselves with emergency operations without stress and time constraints of a functional or major exercise.
technically complete	An ERP that meets all applicable requirements of this directive.
urban centre	A city, town, village, summer village, or hamlet with 50 or more separate buildings, each of which must be an occupied dwelling or any similar development the AER designates as an urban centre.

urban density developments	Any incorporated urban centre, unincorporated rural subdivision, or group of subdivisions with 50 or more separate buildings (which must be occupied dwellings) or any other similar development the AER designates as an urban density development.
water body	Any location where water flows or is present, whether the flow or the presence of water is continuous, intermittent, or occurs only during a flood, including but not limited to aquifers, the bed and shore of a watercourse or wetland, and man-made structures such as a canal, drainage ditch, reservoir, dugout, or other surface feature.
well servicing	Maintenance procedures performed on a producing or injecting well after it has been completed and operations started. Well servicing operations are conducted to maintain or enhance well productivity or injectivity.
workover	The act of re-entering an existing well to perform remedial action to restore or improve productivity or injectivity of the target formation.

Appendix 2 Information on Exposure to Hydrogen Sulphide for the Public in the EPZ

- If you are indoors and can smell hydrogen sulphide (H_2S ; a pungent rotten egg smell), close all the doors and windows and turn down the furnace thermostat immediately. In the summer, turn off all air conditioners.
- Contact the Energy and Environmental Emergency 24-Hour Response Line at 1-800-222-6514 and report the H_2S odour. The AER will immediately act to determine the source of the odour.
- Monitor the radio or television for explanation and instruction. You may also receive a call from the approval holder (company/operator) responsible for the H_2S odour, the authority with jurisdiction in your area, or the AER providing further instruction or explanation.
- If you are considering moving away from the area, contact the approval holder's representative, the authority with jurisdiction in your area, or the AER for evacuation instructions. For major incidents, a special number may be provided and, if so, should be called.
- If environmental monitoring indicates that the H_2S release may affect the public, an evacuation will be carried out under the direction of authorized emergency personnel.
- If you are directed to evacuate, follow the instructions to safely move you away from the H_2S source and out of harm's way. You will also be provided with instructions to check in and register at a reception centre situated in a secure and safe area.
- All other individuals should consider leaving the area and seek medical advice if health symptoms develop.

Energy and Environmental Emergency 24-Hour Response Line: 1-800-222-6514

Appendix 3 Assessment Matrix for Classifying Incidents

The assessment matrix applies to hazards and scenarios described in section 3 of the directive.

2. Consequence of incident			1. Likelihood of incident escalating		
Rank	Category	Example of consequence in category	Rank	Descriptor	Description
1	Minor	- No injuries or public health effects - No environmental effects - Reportable liquid release is contained on site - Gas release effects are on site only - Minor on-site structure or geological feature damage - No or low public or media interest	1	Unlikely	The incident is contained or controlled and is unlikely to escalate. There is no chance of additional hazards. Ongoing monitoring is required.
			2	Moderate	Control of the incident may have deteriorated but imminent control of the hazard by the approval holder is probable. It is unlikely that the incident will escalate.
2	Moderate	- Minor injuries or minor public health effects - Minor environmental effects - Reportable liquid release is <i>not</i> contained on site - Gas release effects may potentially extend beyond the site - Moderate on-site structure or geological feature damage - Potential public or media interest	3	Likely	Imminent or intermittent control of the incident is possible. The approval holder has the capability of using internal and external resources to manage and bring the hazard under control in the near term.
			4	Almost certain or currently occurring	The incident is uncontrolled, and there is little chance that the approval holder will be able to bring the hazard under control in the near term. The approval holder will require assistance from outside parties to remedy the situation.
3	Major	- Injuries requiring hospitalization or potential public health effects require precautionary public protection measure - Liquid spill extends beyond the site—not contained, potential for affecting water bodies and sensitive receptors - Gas release effects extend beyond the site - Major on-site structure or geological feature damage - Public or media interest			
4	Catastrophic	- Multiple injuries, fatalities, or public health effects requiring public protection measures - Liquid spill extends beyond the site—not contained and is affecting water bodies, or sensitive receptors - Gas release effects extend beyond the site - Catastrophic on-site structure or geological feature damage - High public or media interest			

Sum the rank from both columns to obtain the risk level and the incident classification

Assessment results	
Very low 2–3	Alert: An incident that can be handled on site by the approval holder through normal operating procedures and is deemed a very low risk to the public.
Low 4–5	Level 1 emergency: The incident presents no danger outside the approval holder's property or threat to the public and has a minimal environmental impact. Approval holder personnel can manage the incident themselves with immediate control of the hazard. There is little or no media interest.
Medium 6	Level 2 emergency: The incident presents no immediate danger outside the approval holder's property but could potentially extend beyond the approval holder's property. Outside agencies must be notified. Imminent control of the hazard is probable, but there is a moderate threat to the public or the environment or both. There may be local and regional media interest in the event.
High 7–8	Level 3 emergency: The safety of the public is in jeopardy from a major uncontrolled hazard. There are likely significant and ongoing environmental impacts. Immediate multiagency municipal and provincial government involvement is required.

Appendix 4 Information Distributed to the Public at the Onset and During an Incident

To persons evacuated or sheltered at the onset of the incident:

- type and status of the incident
- location and proximity of the incident to people in the vicinity
- public protection measures to follow, evacuation instructions, and any other emergency response measures to consider
- actions being taken to respond to the situation, including anticipated time

To persons evacuated or sheltered during the incident

- description of the products involved and their short-term and long-term effects
- effects the incident may have on people in the vicinity
- areas affected by the incident
- actions the affected public should take if they experience adverse effects

To the public during the incident

- type and status of the incident
- location of the incident
- areas affected by the incident
- description of the products involved
- contacts for additional information
- actions being taken to respond to the situation, including anticipated time

Appendix 5 Assessment and Ignition Criteria Flowchart

